C1.3 Photosynthesis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-c1-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 439–449 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

LIGHT-DEPENDENT REACTIONS

C1.3.9 Photosystems

pp. 439–440

A photosystem is an array of chlorophyll and accessory pigments held by proteins in a membrane: the thylakoids of chloroplasts, or the membranes of cyanobacteria. Pigments absorb light and pass the energy inwards to a **reaction centre** chlorophyll, which emits an excited electron. There are two kinds, photosystem II and photosystem I.

C1.3.10 Why pigments work as an array

p. 440

Different pigments absorb a wider range of wavelengths; many molecules catch more photons; their precise spacing funnels energy to the reaction centre; and the reaction centre passes the excited electron to an acceptor. A single pigment molecule alone would only re-emit the energy as heat or light.

C1.3.11 Photolysis and oxygen

pp. 440–441

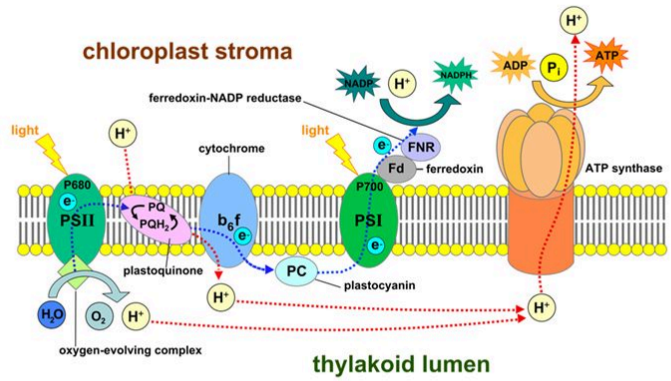
Photosystem II replaces its lost electrons by splitting water: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. The protons and electrons are used; oxygen is waste. Oxygen from early photosynthesizers built up about 2.4 billion years ago, oxidized dissolved iron into banded iron deposits, formed the ozone layer and made aerobic respiration possible.

C1.3.12 Chemiosmosis in thylakoids

pp. 441–443

ATP by chemiosmosis in thylakoids

C1.3.12



Electrons from PSII pass along carriers; the energy **pumps protons** into the thylakoid space.

Protons flow back through **ATP synthase**: **photophosphorylation**.

Electron carriers, proton pumping and ATP synthase. Somepics, CC BY-SA 4.0.

Compare

Same principle as mitochondria; here protons collect in the thylakoid space.

C1.3 Photosynthesis (HL) – Textbook pp. 441–443 11

Excited electrons pass along a chain of carriers that pump protons from the stroma into the thylakoid space. Protons flow back through **ATP synthase**, making ATP. In **non-cyclic** photophosphorylation electrons come from photosystem II (water) and end on NADP; in **cyclic** photophosphorylation electrons from photosystem I return to the chain, making only ATP.

C1.3.13 Reduction of NADP

p. 443

Photosystem I emits excited electrons, two of which reduce one **NADP**, which also takes a proton from the stroma. Use the pairs “NADP and reduced NADP” or “$NADP^+$ and NADPH” consistently.

C1.3.14 The thylakoid as a system

pp. 443–444

Photolysis happens on the inner side of the thylakoid membrane, releasing protons into the thylakoid space. Proton pumping is across the membrane; ATP and reduced NADP are made on the stroma side, where the Calvin cycle uses them.

THE CALVIN CYCLE

C1.3.15 Carbon fixation by rubisco

pp. 444–445

In the stroma, **rubisco** joins CO_2 to **RuBP** (5C), giving two molecules of **glycerate 3-phosphate**.

Rubisco is the most abundant enzyme on Earth; chloroplasts need a lot of it because it works slowly and poorly at low CO_2 concentrations.

C1.3.16 Making triose phosphate

pp. 445–446

Glycerate 3-phosphate is reduced to **triose phosphate**: reduced NADP supplies the hydrogen and ATP supplies the energy.

C1.3.17 Regenerating RuBP

pp. 446–447

Five triose phosphates are converted into three RuBP, using ATP, so the cycle can continue. For each glucose made, five-sixths of all triose phosphate must return to RuBP.

C1.3.18 Making other compounds

pp. 447–448

All the carbon in a plant is fixed in the Calvin cycle. Sugars, starch and cellulose, lipids, and (with nitrogen from nitrate or ammonium) amino acids and nucleotides are made by pathways that start from triose phosphate or other cycle intermediates.

C1.3.19 Interdependence

pp. 448–449

Without light, no ATP or reduced NADP are made, so the Calvin cycle stops within seconds. Without CO_2, ATP and reduced NADP are not used, NADP runs out, electrons back up along the chain and photosystem II stops working.

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